

Chapter 3: Probability Distribution and Probability Density

STAT2920: Introduction to Probability

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- 1 Random Variables
- 2 Probability Distribution
- 3 Continuous Random Variables

- In most applied problems involving probabilities we are interested only in a particular aspect (or in two or a few particular aspects) of the outcomes of experiments.
- For instance, when we roll a pair of dice we are usually interested only in the total, and not in the outcome for each die;
- when we interview a randomly chosen married couple we may be interested in the size of their family and in their joint income, but not in the number of years they have been married or their total assets;
- and when we sample mass-produced light bulbs we may be interested in their durability or their brightness, but not in their price.
- In each of these examples we are interested in numbers that are associated with the outcomes of chance experiments, that is, in the values taken on by **random variables**.

- To be more explicit, consider Figure 1, which pictures the sample space for an experiment in which we roll a pair of dice, and let us assume that each of the 36 possible outcomes has the probability $\frac{1}{36}$.

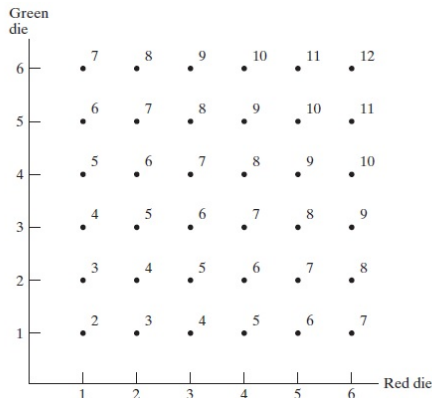


Figure 1: The total number of points rolled with a pair of dice.

Definition 1.

If S is a sample space with a probability measure and X is a real-valued function defined over the elements of S , then X is called a **random variable**.

- In this chapter we shall denote random variables by capital letters and their values by the corresponding lowercase letters;
- for instance, we shall write x to denote a value of the random variable X .

- With reference to the preceding example and Figure 1, observe that the random variable X takes on the value 9, and we write $X = 9$ for the subset

$$\{(6, 3), (5, 4), (4, 5), (3, 6)\}$$

of the sample space S .

Definition 2.

X is a discrete random variable if its range is finite or countable infinite.

Example 1.

Two socks are selected at random and removed in succession from a drawer containing five brown socks and three green socks. List the elements of the sample space, the corresponding probabilities, and the corresponding values w of the random variable W , where W is the number of brown socks selected.

Solution: The solution will be given in class.

Example 2.

A balanced coin is tossed four times. List the elements of the sample space that are presumed to be equally likely, as this is what we mean by a coin being balanced, and the corresponding values x of the random variable X , the total number of heads.

Solution: The solution will be given in class.

Definition 3 (Probability Distribution).

If X is a discrete random variable, the function given by $f(x) = P(X = x)$ for each x within the range of X is called the **probability distribution** of X .

Theorem 1.

A function can serve as the probability distribution of a discrete random variable X if and only if its values, $f(x)$, satisfy the conditions:

1. $f(x) \geq 0$ for each value within its domain;
2. $\sum_x f(x) = 1$, where the summation extends over all the values within its domain.

Example 3.

Find a formula for the probability distribution of the total number of heads obtained in four tosses of a balanced coin.

Solution: The solution will be given in class.

Example 4.

Check whether the function given by

$$f(x) = \frac{x+2}{25} \quad \text{for } x = 1, 2, 3, 4, 5$$

can serve as the probability distribution of a discrete random variable.

Solution: The solution will be given in class.

Example 5.

Check whether the function given by

$$f(x) = \frac{x^2}{30} \quad \text{for } x = 0, 1, 2, 3, 4,$$

can serve as the probability distribution of a discrete random variable.

Solution: The solution will be given in class.

- In some problems it is desirable to present probability distributions graphically, and two kinds of graphical presentations used for this purpose are shown in Figures 2 and 3 .
- The one shown in Figure 2, called a **probability histogram**, represents the probability distribution of Example 3.
- The height of each rectangle equals the probability that X takes on the value that corresponds to the midpoint of its base.

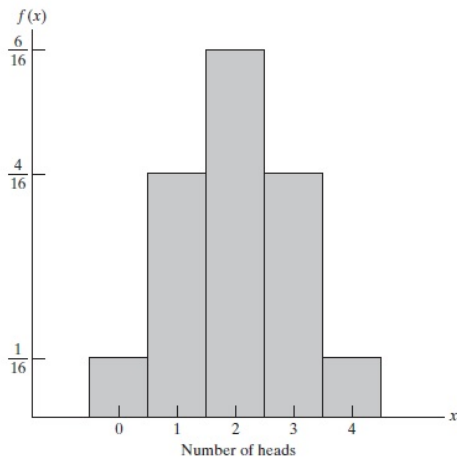


Figure 2: Probability histogram.

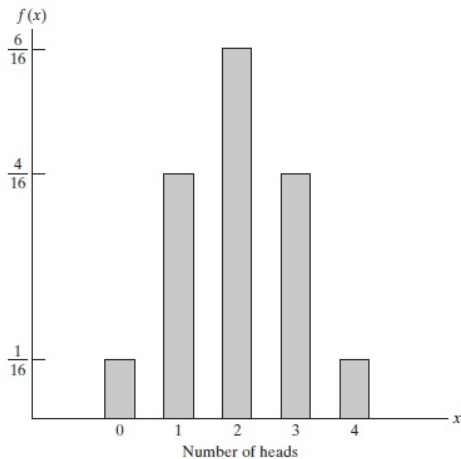


Figure 3: bar chart.

Example 6.

What is the probability of getting less or equal to two heads in tossing a coin three times.

Definition 4 (Distribution function).

If X is a discrete random variable, the function given by

$$F(x) = \mathbb{P}(X \leq x) = \sum_{t \leq x} f(t) \quad \text{for } -\infty < x < \infty \quad (1)$$

where $f(t)$ is the value of the probability distribution of X at t , is called the **distribution function**, or the **cumulative distribution** of X .

- Based on the postulates of probability and some of their immediate consequences, we obtain the following theorem.

Theorem 2.

The values $F(x)$ of the distribution function of a discrete random variable X satisfy the conditions

- $F(-\infty) = 0$ and $F(\infty) = 1$;
- if $a < b$, then $F(a) \leq F(b)$ for any real numbers a and b .

- If we are given the probability distribution of a discrete random variable, the corresponding cumulative distribution function is generally easy to find

Exercise 1.

Let X be the result of a die roll. Plot the cumulative distribution function of X .

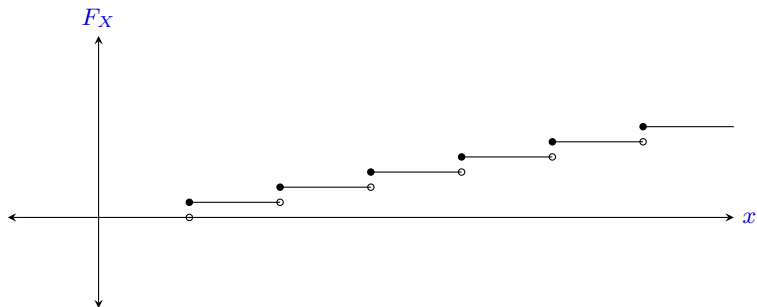


Figure 4: Cumulative Distribution Function for a Die Roll

Exercise 2.

Let X be a random variable whose pmf is given by the following table

x	0	1	Total
$\mathbb{P}_X(x)$	0.5	0.5	1

find the cdf of X .

Example 7.

Find the distribution function of the total number of heads obtained in four tosses of a balanced coin.

Solution: The solution will be given in class.

Example 8.

Two socks are selected at random and removed in succession from a drawer containing five brown socks and three green socks.

1. List the elements of the sample space, the corresponding probabilities, and the corresponding values w of the random variable W , where W is the number of brown socks selected.
2. Find the distribution function of the random variable W and plot its graph.

Solution: The solution will be given in class.

Theorem 3.

If the range of a random variable X consists of the values $x_1 < x_2 < x_3 < \dots < x_n$, then $f(x_1) = F(x_1)$ and

$$f(x_i) = F(x_i) - F(x_{i-1}) \quad \text{for } i = 2, 3, 4, \dots, n$$

Exercise 3.

1. Let

$$f(x) = \begin{cases} \lambda x & x = 1, 2, \dots, 10 \\ 0 & \text{elsewhere,} \end{cases}$$

Find λ such that f is a pmf.

2. Let

$$h(x) = \begin{cases} cx^2 & x = 1, 2, \dots, n \\ 0 & \text{elsewhere,} \end{cases}$$

Find c such that h is a pmf.

Exercise 4.

Find k such that

$$g(x) = \begin{cases} \frac{k2^x}{x!} & \text{for } x = 0, 1, 2, \dots \\ 0 & \text{elsewhere} \end{cases}$$

is a probability function.

Exercise 5.

The probability mass function of a random variable X is given by

$$\mathbb{P}(X = i) = \frac{c\lambda^i}{i!}$$

where $i \in \mathbb{N}$, and $\lambda > 0$.

Find

1. $\mathbb{P}(X = 0)$
2. $\mathbb{P}(X > 2)$

Exercise 6.

Let $h(x) = c(1 - p)^{x-1}$, $x = 1, 2, \dots$, $0 < p < 1$. Find c such that h is a pmf.

For your practice do the following exercises from textbook:

- Exercises 1 to 6 page 72.
- Exercises 7, 9, 11 and 12 page 72.

- Assume that X is a random variable with uncountable sample space;
- It does not make sense to talk about probability attached to each value that X can take;
- In such cases we can only talk of the probability that X will fall in an interval, for instance

$$\mathbb{P}(a \leq X \leq b),$$

for some a, b real numbers.

Our specific goal

- Finding the probability that X falls in some interval, that is finding $\mathbb{P}(a < X < b)$, where a and b are some constants.
- We'll do this by using $f(x)$, the probability density function (*pdf*) of X , and $F(x)$, the cumulative distribution function (*cdf*) of X .

Definition 5.

A function with values $f(x)$, defined over the set of all real numbers, is called a probability density function of the continuous random variable X if and only if

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx$$

for any real constants a and b with $a \leq b$.

- Probability density function is written as **pdf**.

Theorem 4.

If X is a continuous random variable and a and b are real constants with $a \leq b$, then

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X < b).$$

Theorem 5.

A function can serve as a probability density of a continuous random variable X if its values, $f(x)$, satisfy the conditions

1. $f(x) \geq 0$, for $-\infty < x < \infty$
2. $\int_{-\infty}^{\infty} f(x) dx = 1$.

Example 9.

Let X has the probability density

$$f(x) = \begin{cases} ke^{-3x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find k and $\mathbb{P}(0.5 \leq X \leq 1)$.

Definition 6.

If X is a continuous random variable and the value of its probability density at t is $f(t)$, then the function given by

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt \quad \text{for } -\infty < x < \infty.$$

Theorem 6.

If $f(x)$ and $F(x)$ are the values of the probability density and the distribution function of X at x , then

$$\mathbb{P}(a \leq X \leq b) = F(b) - F(a)$$

for any real constants a and b with $a \leq b$ and

$$f(x) = \frac{dF(x)}{dx}$$

where the derivative exists.

Exercise 7.

Find the cdf of X if its pdf is

$$f(x) = \begin{cases} 3e^{-3x} & \text{if } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Exercise 8.

If X has cdf given by

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{x+1}{2} & \text{if } -1 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Find $\mathbb{P}\left(-\frac{1}{2} < X < \frac{1}{2}\right)$

Exercise 9.

Suppose the pdf of a continuous random variable X is defined as:

$$f(x) = \begin{cases} x + 1 & \text{if } -1 < x < 0 \\ 1 - x & \text{if } 0 \leq x < 1 \end{cases}$$

Find and graph the cdf. $F(x)$.

Exercise 10.

The actual amount of coffee (in grams) in a 230-gram jar filled by a certain machine is a random variable whose probability density is given by

$$f(x) = \begin{cases} 0 & \text{for } x \leq 227.5 \\ \frac{1}{5} & \text{for } 227.5 < x < 232.5 \\ 0 & \text{for } x \geq 232.5 \end{cases}$$

Find the probabilities that a 230-gram jar filled by this machine will contain

1. at most 228.65 grams of coffee;
2. anywhere from 229.34 to 231.66 grams of coffee;
3. at least 229.85 grams of coffee.

Exercise 11.

The number of minutes that a flight from Phoenix to Tucson is early or late is a random variable whose probability density is given by

$$f(x) = \begin{cases} \frac{1}{288} (36 - x^2) & \text{for } -6 < x < 6 \\ 0 & \text{elsewhere} \end{cases}$$

where negative values are indicative of the flight's being early and positive values are indicative of its being late. Find the probabilities that one of these flights will be

1. at least 2 minutes early;
2. at least 1 minute late;
3. anywhere from 1 to 3 minutes early;
4. exactly 5 minutes late.

Exercise 12.

The total lifetime (in years) of five-year-old dogs of a certain breed is a random variable whose distribution function is given by

$$F(x) = \begin{cases} 0 & \text{for } x \leq 5 \\ 1 - \frac{25}{x^2} & \text{for } x > 5 \end{cases}$$

Find the probabilities that such a five-year-old dog will live

1. beyond 10 years;
2. less than eight years;
3. anywhere from 12 to 15 years.

Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

Some useful websites:

- Society of Actuaries, [Exam of probability, sample questions](#).
- Society of Actuaries, [Exam of probability, sample solutions](#).